

UNIX Study Material

Imp programmes for Unix

1. Check Even or Odd Number

```
echo "Enter a number:"
read n

r=`expr $n % 2`

if [ $r -eq 0 ]
then
    echo "Even Number"
else
    echo "Odd Number"
fi
```

2. Check Whether a Number is Positive or Not

```
echo "Enter a number:"
read n

if [ $n -gt 0 ]
then
    echo "Positive Number"
else
    echo "Not Positive"
fi
```

3. Check Whether a Year is Leap Year or Not

```
echo "Enter Year:"
read y
```

```
r=`expr $y % 4`  
  
if [ $r -eq 0 ]  
then  
    echo "Leap Year"  
else  
    echo "Not Leap Year"  
fi
```

4. Check Whether a Person is Senior Citizen or Not

```
echo "Enter Age:"  
read age  
  
if [ $age -ge 60 ]  
then  
    echo "Senior Citizen"  
else  
    echo "Not Senior Citizen"  
fi
```

5. Calculate Road Tax

```
echo "Enter Bike Price:"  
read p  
  
if [ $p -gt 100000 ]  
then  
    tax=`expr $p \* 15 / 100`  
elif [ $p -gt 50000 ]  
then  
    tax=`expr $p \* 10 / 100`  
else  
    tax=`expr $p \* 5 / 100`  
fi
```

```
echo "Road Tax = Rs. $tax"
```

6. Calculate Electric Bill

```
echo "Enter Units:"
read u

if [ $u -le 100 ]
then
    bill=0
elif [ $u -le 200 ]
then
    x=`expr $u - 100`
    bill=`expr $x \* 5`
else
    x=`expr $u - 200`
    bill=`expr 100 \* 5 + $x \* 10`
fi

echo "Bill Amount = Rs. $bill"
```

7. Check Whether a Number is Three Digit or Not

```
echo "Enter Number:"
read n

if [ $n -ge 100 -a $n -le 999 ]
then
    echo "Three Digit Number"
else
    echo "Not Three Digit Number"
fi
```

8. Menu Based Calculator

```
echo "Enter Two Numbers:"
read a b

echo "1.Add"
echo "2.Subtract"
echo "3.Multiply"
echo "4.Divide"

read ch

case $ch in
1) echo `expr $a + $b` ;;
2) echo `expr $a - $b` ;;
3) echo `expr $a \* $b` ;;
4) echo `expr $a / $b` ;;
*) echo "Invalid Choice" ;;
esac
```

9. Sum of First N Natural Numbers

```
echo "Enter N:"
read n

i=1
sum=0

while [ $i -le $n ]
do
    sum=`expr $sum + $i`
    i=`expr $i + 1`
done

echo "Sum = $sum"
```

10. Check Palindrome Number

```

echo "Enter Number:"
read n

temp=$n
rev=0

while [ $n -gt 0 ]
do
    d=`expr $n % 10`
    rev=`expr $rev \* 10 + $d`
    n=`expr $n / 10`
done

if [ $temp -eq $rev ]
then
    echo "Palindrome"
else
    echo "Not Palindrome"
fi

```

11. Check Armstrong Number (3 Digit)

```

echo "Enter Number:"
read n

temp=$n
sum=0

while [ $n -gt 0 ]
do
    d=`expr $n % 10`
    sum=`expr $sum + $d \* $d \* $d`
    n=`expr $n / 10`
done

if [ $temp -eq $sum ]
then

```

```
        echo "Armstrong Number"
    else
        echo "Not Armstrong Number"
    fi
```

12. Count Total Even and Odd Digits

```
echo "Enter Number:"
read n

even=0
odd=0

while [ $n -gt 0 ]
do
    d=`expr $n % 10`
    r=`expr $d % 2`
    if [ $r -eq 0 ]
    then
        even=`expr $even + 1`
    else
        odd=`expr $odd + 1`
    fi
    n=`expr $n / 10`
done

echo "Even Digits = $even"
echo "Odd Digits = $odd"
```

Imp command for Unix

1. Check if File has .sh Extension

```
echo "Enter File Name:"
read file
```

```
case "$file" in
    *.sh) echo "Shell Script File" ;;
    *) echo "Not a Shell Script File" ;;
esac

# Example Input: test.sh -> Output: Shell Script File
```

2. Show Files with 3 Characters Names

```
ls ???

# Example:
# Files: abc, xyz, test, hello
# Output: abc, xyz
```

3. Show Files Starting with a, b, or c

```
ls [abc]*

# Example:
# Files: apple, ball, cat, dog, book
# Output: apple, ball, book, cook
```

UNIX File System Structure

Memory Trick:

- **Directory** = Filename (Mapping)
- **Inode** = Information / Metadata (Type, Size, Owner, etc.)
- **Disk Block** = Actual Content

Component	Description / Contents
Directory	Stores Filename and Inode Number link.

Inode	Stores Type, Owner, Permissions, Size, Timestamps, and Block Locations. (Does NOT store filename).
Disk Blocks	Stores the actual data/content of the file.

Flow:

Directory → Filename → Inode → Disk Blocks → Actual Data